# Audio Deep Learning

# Assignment #3

# Automatic Speech Recognition

## Assignment overview and motivation

- *Format: Individual or group (max 3 students) with offline defense*
- *Grading: 17 points*
- *Timeline: Assignment # 3 available on 16th June, Code submission deadline till 5th July, and offline defense deadline till 19th July*
- ***Submission via Google Form**: https://forms.gle/zvtsJJYWSXbRUZc97*

*While completing this assignment, you will develop and strengthen the following skills:*

- *Approaching a real-world Automatic Speech Recognition (ASR) problem using naturalistic speech recordings and evaluating system performance with standard recognition metrics such as Word Error Rate (WER) and Character Error Rate (CER).*
- *Conducting effective Exploratory Data Analysis (EDA) on speech recordings, transcripts, and acoustic characteristics (e.g., duration distributions, speaking rates, signal quality, and vocabulary statistics) to better understand the dataset and its challenges.*
- *Designing a reliable speaker-independent validation strategy that prevents data leakage between training and validation sets while ensuring robust estimation of model performance.*
- *Using Self-Supervised Learning (SSL) speech models (such as wav2vec 2.0, HuBERT, or Whisper encoders) as pretrained representations for Automatic Speech Recognition tasks.*
- *Performing a comprehensive ablation study on fine-tuning strategies, decoding approaches, language model integration, data augmentation techniques, and training configurations to identify the most effective ASR pipeline.*

## Task

*Task Selection:*

- *Perform Speech Recognition Task on Toronto Dataset*
- *Perform Phoneme Recognition on the TIMIT dataset*

*EDA and Metric Analysis:*

- *Perform Exploratory Data Analysis (EDA) on both datasets*

- *Analyze the proposed target metrics: Character Error Rate (CER) and Word Error Rate (WER):*
- *Suggest complementary metrics to capture other aspects of performance.*
- *Propose an alternative target metric, with justification.*

*Validation:*

- *Use next test files: [ 'toronto_27', 'toronto_46', 'toronto_42', 'toronto_37', 'toronto_89', 'toronto_43', 'toronto_157', 'toronto_9', 'toronto_156', 'toronto_7', 'toronto_123', 'toronto_54', 'toronto_67', 'toronto_62', 'toronto_81', 'toronto_134', 'toronto_148', 'toronto_21', 'toronto_135', 'toronto_166', 'toronto_58' ]*
- *Design a separate validation split from the train set*
- *Explain your motivation for the chosen strategy and the importance of balancing speaker distributions, utterance lengths, and transcript complexity within the splits.*
- *Discuss the challenges of achieving representative train/validation distributions in naturalistic speech datasets.*

*Whisper Approach:*

- *Select an appropriate pretrained Whisper checkpoint and justify your choice.*
- *Implement the training pipeline using PyTorch Lightning, including data loading, preprocessing, optimization, and metric computation.*
- *Fine-tune Whisper on the Toronto Dataset for Speech Recognition and on the TIMIT dataset for Phoneme Recognition.*
- *Adapt the output vocabulary, tokenization strategy, and decoding procedure to the requirements of each task.*
- *Collect all relevant results:*
    - *metrics (CER and WER for Speech Recognition; PER for Phoneme Recognition);*
    - *inference examples and hard samples analysis (e.g., rare words, proper names, noisy recordings, confusable phonemes, and long utterances).*
- *Conclude whether the chosen Whisper model is adequate for both tasks.*
- *Prepare an ablation study on the tested models and training configurations.*

*Self-Supervised Approach (Data2Vec, Wave2Vec, …):*

- *Explore pretrained self-supervised speech encoders for both Speech Recognition and Phoneme Recognition tasks.*
- *Build ASR and phoneme-recognition systems using the selected encoder together with an appropriate prediction head (e.g., CTC-based recognition).*
- *Fine-tune the selected models on the Toronto Dataset for Speech Recognition and on the TIMIT dataset for Phoneme Recognition.*
- *Collect all relevant results:*
    - *metrics (CER and WER for Speech Recognition; PER and optionally CER for Phoneme Recognition);*
    - *inference examples and hard samples analysis.*
- *Conclude whether the model is adequate for the corresponding task.*
- *Prepare an ablation study on the tested models, exploring freezing depths (e.g., number of frozen transformer blocks), decoding strategies, loss functions, audio augmentation techniques, and model size.*

*Audio Filtering and Signal Enhancement:*

- *Explore audio preprocessing and filtering techniques aimed at improving Speech Recognition and Phoneme Recognition performance.*

- *Investigate approaches such as noise reduction, band-pass filtering, silence trimming, volume normalization, voice activity detection (VAD), dereverberation, and sample-rate conversion.*
- *Analyze how different preprocessing pipelines affect both audio quality and downstream model performance.*
- *Collect all relevant results:*
    - *metrics (CER and WER for Speech Recognition; PER for Phoneme Recognition);*
    - *qualitative analysis of challenging samples before and after filtering;*
    - *examples where preprocessing improves or degrades recognition quality.*
- *Conclude whether audio filtering provides a meaningful performance gain for the selected datasets and models.*
- *Prepare an ablation study comparing individual filtering methods and combinations of preprocessing techniques.*

*Final Report*

- *Prepare a comprehensive report including:*
    - *data insights,*
    - *feature engineering,*
    - *tried algorithms and models,*
    - *chosen hyperparameters,*
    - *metrics,*
    - *insights and conclusions.*
- *Reports may be prepared in LaTeX or Markdown (README.md).*

***NOTES***

- *Do not hesitate to include failed ideas or experiments in your report. Documenting unsuccessful attempts is a strong indication that you systematically explored different approaches and gradually arrived at the best-performing solution, rather than relying on a "lucky shot." Moreover, analyzing failed approaches can often lead to valuable insights about the dataset, the task itself, and the behavior of deep learning algorithms. Such reflections may also highlight potential directions for future improvements and demonstrate a deeper understanding of the problem-solving process.*
- *Keep your EDA concise. Avoid overloading it with excessive plots, histograms, or tables. Focus only on key aspects of the data (e.g., missing values, target distribution). If a part of your EDA does not lead to meaningful conclusions, you likely don't need it (for example, full pairwise correlation matrices). Do NOT follow the typical Kaggle-style EDA — most of them are far too verbose and unfocused.*
- *Prepare your presentation carefully. A messy speech with too much code or too many files makes it nearly impossible for the audience to follow. Think in advance about what you want to highlight*

# Grading Policy and Defense

*Grading Policy is outlined in the table*

| ***Criterion*** | ***Points*** |
|---|---|
| *Overall correctness and completeness of the submitted code for the lab assignment, overall quality (metrics) of the proposed solution* | *5* |
| *Practical question about the submitted code during defense* | *7* |
| *Theoretical part during defense (2 questions + 1 exercise)* | *5* |
| ***Criterion*** | ***Points*** |

*The defense will be conducted during a practical session, where the Lecturer will evaluate your code, report, and presentation skills. It is important to emphasize that the assessment will not be limited to metrics, analysis, and the performance of your models, but will also take into account the overall quality of your code, including readability, structure, and adherence to good practices. During the defense, the Lecturer will evaluate each student's individual contribution, demonstrated skills, and depth of knowledge. In addition, up to 2 theoretical questions may be asked; these will be based on the questions outlined in Appendix 1, but will not necessarily be restricted to them.*

***Exercise with practical tasks similar to Appendix 2 will be evaluated in the form of a written test on one of the practical sessions***

## AI Tools usage policy

***Scope***

*This policy applies to all generative AI tools that produce or modify text, code, images, or other digital content. This includes, but is not limited to:*

- *Text generation tools: ChatGPT, Claude, Gemini, Mistral, LLaMA-based models, etc.*
- *Code generation tools: GitHub Copilot, OpenAI Codex, TabNine, etc.*
- *Language assistance tools: Grammarly, DeepL, Google Translate, etc.*
- *Local LLMs or self-hosted AI models.*

***Allowed Uses***

*You may use AI tools for the following purposes, provided they comply with the conditions below:*

- *Spell Check & Style Correction. Example: Using Grammarly to correct grammar and improve style in a lab report or paper review.*
- *Translation. Example: Using DeepL or Google Translate to translate sections of a document into another language.*

- *Idea Generation & Pitching. Example: Brainstorming possible approaches for solving a problem or generating research ideas.*
- *Self-Education. Example: Using ChatGPT to explain an unfamiliar algorithm or concept.*
- *Library & Documentation Exploration. Example: Asking AI to summarize PyTorch documentation for a specific class or function.*
- *Code Assistance (Partial). Writing or correcting specific functions or methods (not entire solutions). Examples: Write a function that reads data and converts it to another format. Implement a data collator with padding to the longest sequence. Create an image augmentation function for training.*

***Conditions for Use***

*Full Transparency:*

- *Any use of AI tools must be disclosed in your submission.*
- *Include all prompts, responses, and/or screenshots showing your interaction with the AI tool.*
- *If using online tools, include chat links (if available) or full transcripts in an appendix of your report.*

*Partial Contribution Only:*

- *AI tools should support your work — they must not produce the complete assignment, lab, or report.*
- *You must be able to explain and defend any AI-generated content during oral defense or Q&A*

***Examples***

*Acceptable Uses:*

- *"Correct grammar and style in this paragraph of my report."*
- *"Write a function to load images from a directory and resize them."*
- *"Suggest 3 ideas for improving model generalization."*

*Unacceptable Uses:*

- *"Write the entire peer review for this paper."*
- *"Generate the complete solution code for this assignment task."*

# Appendix 1 (Theoretical Questions)

- *Explain the difference between Automatic Speech Recognition (ASR) and Phoneme Recognition. Why is Phoneme Error Rate (PER) commonly used for the latter, while Word Error Rate (WER) is preferred for ASR?*
- *What are the strengths and weaknesses of Character Error Rate (CER) and Word Error Rate (WER)? In which scenarios can CER provide a more informative evaluation than WER?*
- *Explain how insertions, deletions, and substitutions contribute to WER. Provide an example where two systems achieve similar WER values but make qualitatively different mistakes.*
- *What are the primary challenges in building robust ASR systems for spontaneous, real-world speech compared to read speech datasets?*
- *Why is speaker-independent evaluation important in ASR? What forms of data leakage can artificially inflate recognition performance?*

- *Explain the role of text normalization in ASR evaluation. Why might punctuation, capitalization, numbers, and abbreviations significantly affect reported metrics?*
- *What are the advantages and disadvantages of subword tokenization methods (e.g., BPE or SentencePiece) compared to character-level tokenization in ASR systems?*
- *Explain the core idea behind self-supervised learning (SSL) in speech processing. Why are SSL representations useful for low-resource speech tasks?*
- *What is the difference between contrastive masked prediction (e.g., wav2vec 2.0) and predictive self-distillation (e.g., data2vec)? What information is each objective attempting to learn?*
- *Explain how wav2vec 2.0 is trained. What are the roles of masking, quantization, and the contrastive objective?*
- *What is the purpose of the CTC (Connectionist Temporal Classification) loss function? Why is it particularly useful for speech recognition?*
- *Explain the blank token in CTC. Why is it necessary, and how does it enable alignment between audio frames and transcription targets?*
- *Describe the CTC decoding process. How does a sequence of frame-level predictions get converted into a final transcription?*
- *Compare greedy decoding and beam search decoding. Why can beam search improve ASR performance even when the acoustic model remains unchanged?*
- *What are the main limitations of CTC-based ASR systems compared to encoder-decoder approaches?*
- *Explain the encoder-decoder paradigm used by Whisper. How does it differ from a CTC-based architecture?*
- *What is teacher forcing, and why is it commonly used during training of autoregressive ASR models?*
- *Explain the attention mechanism in sequence-to-sequence models. Why is attention particularly important for long audio recordings?*
- *What is the difference between self-attention and cross-attention in Transformer architectures?*
- *In Whisper, what information is processed by the encoder and what information is processed by the decoder?*
- *Why can multilingual ASR models outperform monolingual models in some low-resource languages?*
- *Explain the difference between zero-shot inference, feature extraction, partial fine-tuning, and full fine-tuning of pretrained speech models.*
- *When might freezing a large portion of a pretrained SSL encoder be beneficial? When can it hurt performance?*
- *What are the potential failure modes of transfer learning in speech recognition?*
- *Explain how Voice Activity Detection (VAD) can improve an ASR pipeline. Under what conditions might VAD degrade performance?*
- *Why are audio preprocessing techniques such as noise reduction, loudness normalization, and silence trimming often applied before ASR training or inference?*
- *What types of audio augmentations are commonly used for ASR (e.g., noise injection, reverberation, speed perturbation)? Why are these augmentations generally safer than pitch-shifting for phoneme recognition tasks?*
- *What are the main differences between read-speech datasets (e.g., TIMIT) and spontaneous speech datasets? How do these differences affect model development and evaluation?*
- *Why is phoneme recognition often considered a useful intermediate task for studying speech representations?*

- *Explain the trade-offs between Whisper-style encoder-decoder models and SSL + CTC approaches (e.g., wav2vec 2.0 + CTC) in terms of accuracy, computational cost, decoding complexity, and adaptability to new languages.*

# Appendix 2 (Practical Exercises)

# Audio ML Lab №3 — Practical Exercises

## Exercise 1: Connectionist Temporal Classification (CTC) Loss

Consider the following vocabulary:

$$\{_, A, B\}$$

where _ denotes the CTC blank token.

Suppose a model produces predictions for $T = 3$ time steps:

$$P = \begin{array}{c|ccc} & _ & A & B \\ \hline t=1 & 0.1 & 0.6 & 0.3 \\ t=2 & 0.7 & 0.2 & 0.1 \\ t=3 & 0.2 & 0.2 & 0.6 \end{array}$$

The target sequence is:

$$Y = [A, B]$$

Assume that the CTC loss is computed as:

$$\mathcal{L}_{CTC} = -\log P(Y|X)$$

where $P(Y|X)$ is obtained by summing the probabilities of all valid alignments that collapse to the target sequence $[A, B]$.

### Tasks

1. List all valid CTC alignments that collapse to the target sequence $[A, B]$.
2. Compute the probability of each alignment.
3. Compute $P(Y|X)$.
4. Compute the CTC loss $\mathcal{L}_{CTC}$.

## Exercise 2: Word Error Rate (WER) and Character Error Rate (CER)

Consider the following transcription:

Ground Truth: `"I LIKE AUDIO"`

and the model prediction:

$$\text{Prediction: } \texttt{"I LOVE AUDO"}$$

Recall that

$$WER = \frac{S + D + I}{N_w}$$

where:

- $S$ — number of word substitutions,
- $D$ — number of word deletions,
- $I$ — number of word insertions,
- $N_w$ — number of words in the reference.

Similarly,

$$CER = \frac{S_c + D_c + I_c}{N_c}$$

where the same quantities are computed at the character level.

Ignore spaces when computing CER.

**Tasks**

1. Determine the minimum edit sequence (substitutions, insertions, deletions) at the word level.
2. Compute the Word Error Rate (WER).
3. Determine the minimum edit sequence (substitutions, insertions, deletions) at the character level.
4. Compute the Character Error Rate (CER).

## Exercise 3: Beam Search Decoding

Consider the following frame-level token probabilities produced by a model.

Vocabulary:

$$\{A, B, \langle eos \rangle\}$$

Beam width:

$$K = 2$$

Predicted probabilities:

| | $A$ | $B$ | $\langle eos \rangle$ |
|---|---|---|---|
| $t = 1$ | 0.6 | 0.4 | 0.0 |
| $t = 2$ | 0.3 | 0.5 | 0.2 |
| $t = 3$ | 0.2 | 0.2 | 0.6 |

At each step, Beam Search keeps only the top-$K$ partial hypotheses according to their cumulative probability.

**Tasks**

1. Perform Beam Search decoding step-by-step using beam width $K = 2$.
2. List the surviving hypotheses after each decoding step.
3. Determine the final decoded sequence.
4. Compare the result with Greedy Decoding.

## Exercise 4: Variational Autoencoder (VAE) Loss

The VAE objective consists of two terms:

$$\mathcal{L}_{VAE} = \mathcal{L}_{rec} + \mathcal{L}_{KL}$$

where:

- $\mathcal{L}_{rec}$ is the reconstruction loss,
- $\mathcal{L}_{KL}$ is the KL-divergence between the approximate posterior $q(z|x)$ and the standard normal prior $p(z) = \mathcal{N}(0, I)$.

Assume a simplified one-dimensional input:

$$x = [1.0, 0.0], \qquad \hat{x} = [0.8, 0.3]$$

Use Mean Squared Error (MSE) as the reconstruction loss:

$$\mathcal{L}_{rec} = \frac{1}{N}\sum_{i=1}^{N}(x_i - \hat{x}_i)^2$$

The encoder predicts the following latent distribution parameters:

$$\mu = [0.5, -0.5]$$

$$\log \sigma^2 = [0.0,\ \log(0.25)]$$

For a diagonal Gaussian posterior, the KL term is:

$$\mathcal{L}_{KL} = -\frac{1}{2}\sum_{j=1}^{d}\left(1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2\right)$$

**Tasks**

1. Compute the reconstruction loss $\mathcal{L}_{rec}$.
2. Compute the KL-divergence term $\mathcal{L}_{KL}$.
3. Compute the total VAE loss $\mathcal{L}_{VAE}$.